# REDCap advanced features

## Quick reference: field embedding, piping, action tags, smart variables, special functions

Supplementary material for REDCap Part 2, Research Data Management, MEPI, University of Antwerp, 2026-27. Examples come from TBCARE (class demo) and TUMAINI (homework). The full, current help pages are inside REDCap: in the Online Designer, click Action Tags, Piping, Smart Variables, Field Embedding or Special Functions.

## 1 Field embedding

Moves a field to a place you choose on the same form, for example into a table. Write the variable name in **curly brackets** in the label, field note, section header or choice label of another field.

| You write (label of a descriptive field) | Result |
|---|---|
| `<table><tr><th>Weight</th><th>Height</th></tr>` `<tr><td>{weight_kg}</td><td>{height_cm}</td></tr></table>` | The two input boxes appear side by side in a table. |
| `I agree to take part {cons_part}` | The Yes / No buttons appear after the sentence (TBCARE e-Consent). |
| `{known_n} {known_unit}` | Number and unit on one line (TUMAINI FSX 1g). |

- Same instrument only; on a multi-page survey, the same page.
- The embedded field keeps its validation, required status and branching logic. Its own label is not shown.
- Embed a field only once. Test it on a real record: the preview in the Online Designer is only approximate.

## 2 Piping

Shows an answer that was already given, inside new text. Write the variable name in **square brackets**.

| You write | REDCap shows |
|---|---|
| `Of these [sex_n] times, how many with a condom?` | Of these 6 times, how many with a condom? |
| `[sex]` and `[sex:value]` | The label (Female) and the code (2) of a multiple-choice answer |
| `[contra_m:checked]` | The ticked options of a checkbox field |
| `[baseline_arm_1][weight_kg]` | The value at a named event |
| `[first-event-name][first_name]` | The value at the first event of the record's arm (works in every arm) |
| `[ae_term]` in a repeating form | The value of the same instance |

Piping works in labels, field notes, section headers, choice labels, survey instructions and completion text, survey invitations, alerts, custom record labels, and the `@DEFAULT` action tag.

## 3 Action tags

Write them in the **Field Annotation** of a field. Several can be combined, separated by a space.

| Action tag | What it does | Example |
|---|---|---|
| `@TODAY @NOW` | Fills in today's date or the date and time | Visit date |
| `@HIDDEN @HIDDEN-SURVEY @HIDDEN-FORM` | Hides the field everywhere, on surveys only, or on forms only | A date staff need, participants not |
| `@READONLY` | Shows the field but it cannot be changed | A calculated text |
| `@CHARLIMIT=n` | At most n characters, with a counter | Initials (4), phone (16) |
| `@CALCTEXT(…)` | A calculation that returns text | `@CALCTEXT(if([eligible] = 1, 'ELIGIBLE', 'NOT ELIGIBLE'))` |
| `@CALCDATE(…)` | A date calculated from another date | Next visit 91 days after Day 0 |
| `@NONEOFTHEABOVE='6'` | Choosing code 6 unticks all other options | Symptoms: None |
| `@HIDECHOICE='3'` | Hides an option but keeps data already entered with it | An answer option you retired |
| `@DEFAULT='…'` | A starting value; can pipe | `@DEFAULT='[record-dag-name]'` |
| `@LATITUDE @LONGITUDE` | Captures the GPS position of the device | Home position (LOC) |
| `@WORDLIMIT=n` | At most n words | Free-text comments |

## 4 Smart variables

Filled in by REDCap from the context. Use them in logic, calculations and piping.

| Smart variable | Gives |
|---|---|
| `[record-name]` | The record ID |
| `[record-dag-name] [record-dag-label]` | The record's Data Access Group (unique name or label) |
| `[event-name] [event-label]` | The current event (unique name or label) |
| `[first-event-name] [previous-event-name] [next-event-name]` | Events relative to the current one, within the record's arm |
| `[arm-number] [arm-label]` | The arm of the record |
| `[user-name] [user-dag-name]` | The user who is logged in and their DAG |
| `[current-instance]` | The instance number in a repeating form or event |
| `[survey-link:fsx]` | A link to a survey |

Example from TUMAINI: hide the speculum questions in the Reduced arm with `[fge_done] = '1' and [arm-number] <> '3'`.

## 5 Special functions

| Function | Example |
|---|---|
| `datediff(d1, d2, unit)` | `rounddown(datediff([dob], [scr_date], 'y'), 0)`: age in whole years |
| `if(condition, yes, no)` | `if([qtcf] > 500, 1, 0)` |
| `round() rounddown() roundup()` | `round([weight_kg] / ([height_cm] / 100) ^ 2, 1)`: BMI |
| `sum() mean() min() max()` | `sum([qol_1], [qol_2], [qol_3], [qol_4])` (blanks are ignored) |
| `isblankormissingcode()` | `isblankormissingcode([dob])`: true if empty |
| `concat() left() upper()` | `upper(left([last_name], 2))` |
| `contains() starts_with()` | `contains([ae_term], 'rash')` |

**Before you trust a calculation**, make a test record where you know the answer, and check what happens when a value is missing.